CORVINUS UNIVERSITY of BUDAPEST

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Capital Structure II

International Corporate Finance – 5

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Based on:
Berk – DeMarzo (2017): Corporate Finance. Fourth edition, Pearson – chapter 16

Optimal capital structure

<u>an optimal capital structure may exist in the presence of</u>:

1. **Bankruptcy costs**
2. **Signalling effects**
3. **Agency costs**

Bankruptcy costs

MM assume zero bankruptcy costs (BC)
⇒ as soon as V<D, bondholders get V

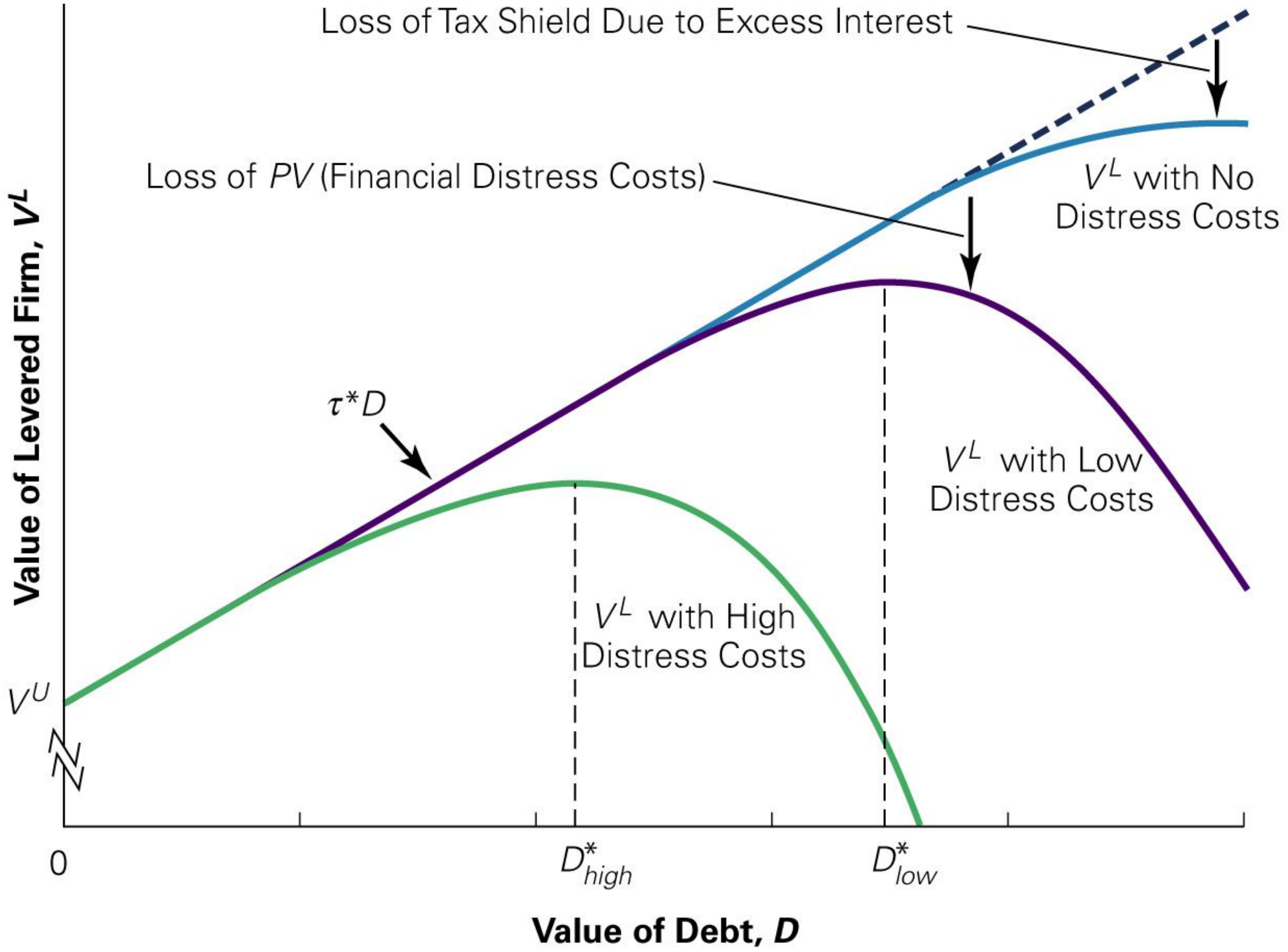
BUT, if BC>0, as V<D bondholders get only V-BC

This means that cost of debt will increase with leverage as bondholders perceive an increasing risk of bankruptcy

Bankruptcy costs

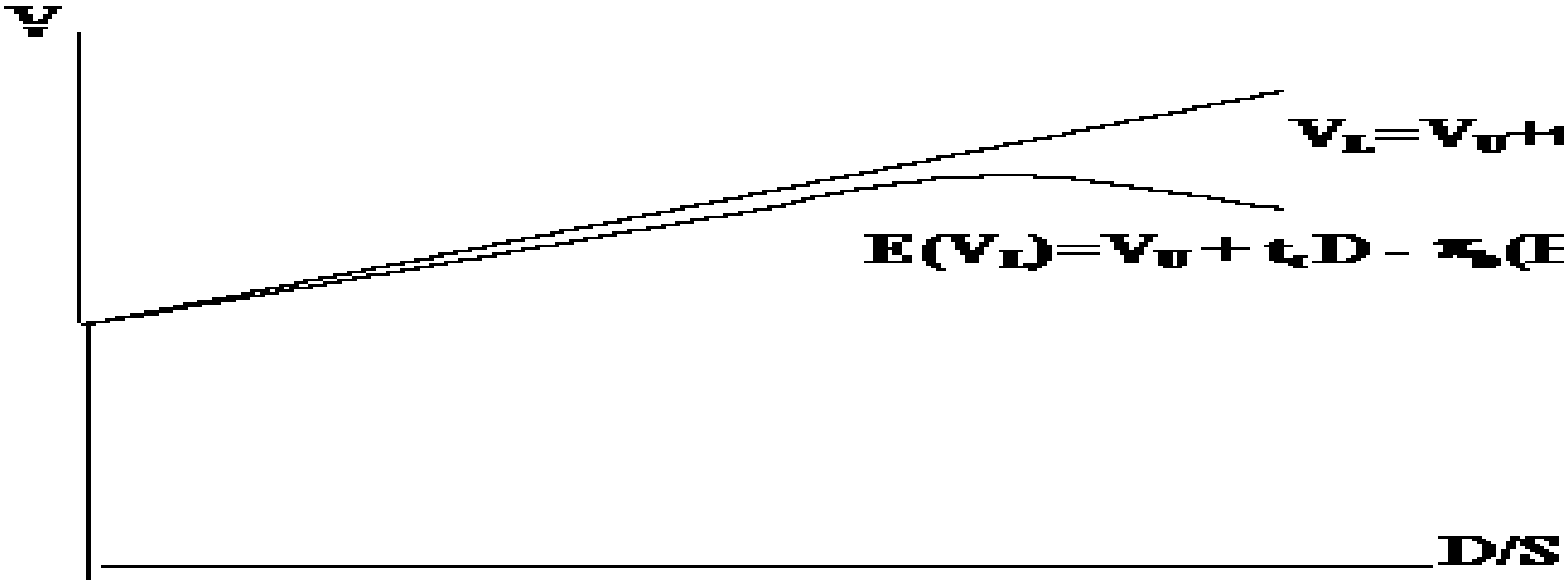
- Average direct costs: 3% to 4% of the pre-bankruptcy market value of total assets
- Indirect costs: 10% to 20% of firm value
    - Loss of Customers
    - Loss of Suppliers
    - Loss of Employees
    - Loss of Receivables
    - Fire Sale of Assets
    - Delayed Liquidation
    - Costs to Creditors

Optimal Leverage with Taxes and Financial Distress Costs



Bankruptcy costs

Thus $\Rightarrow E(V_L)=V_U + t_cD - \pi_b(BC)$
where π_b= probability of bankruptcy



More Recent Literature: Agency, Signalling and More

1. Improve conflicts of interest among interest groups (agency)

2. Convey private information to outsiders, or mitigate adverse selection (asymmetric information)

3. Improve the outcome of corporate control contests

Agency costs of borrowing

- Derive from the conflict of interest between equity investors and lenders
- Equity investors have a residual claim on cash flows $\Rightarrow$ accept any action that increases firm's value, even if it also increases bankruptcy risk
- Bondholders have a fixed claim on cash flows $\Rightarrow$ maximise the security of their claim
- This conflict affects decisions on:
  - (i) Investment
  - (ii) How to finance it, and
  - (iii) Dividend distribution

Excessive risk-taking and asset substitution

- When a firm faces financial distress, shareholders can gain at the expense of debt holders by taking a negative-NPV project, if it is sufficiently risky: **Excessive risk-taking and asset substitution**
- **When a firm faces financial distress, it may choose not to finance new, positive-NPV projects : Debt overhang**

Agency costs of borrowing: Excessive risk-taking and asset substitution

- Baxter has a loan of $1 million due at the end of the year
- Without a change in its strategy, the market value of its assets will be only $900,000 at that time, and Baxter will default on its debt
- The new strategy requires no upfront investment, but it has only a 50% chance of success
- If the new strategy succeeds, it will increase the value of the firm's asset to $1.3 million
- If the new strategy fails, the value of the firm's assets will fall to $300,000
- Can shareholders benefit from the strategy?

Agency costs of borrowing: Excessive risk-taking and asset substitution

		New Risky Strategy		
	Old Strategy	Success	Failure	Expected
Value of assets	**900**	1300	300	**800**
Debt	**900**	1000	300	**650**
Equity	**0**	300	0	**150**

Agency costs of borrowing: Debt Overhang and Under-Investment

- Investment opportunity that requires an initial investment of $100,000 and will generate a risk-free return of 50%
- If the current risk-free rate is 5%, this investment clearly has a positive NPV
  - What if Baxter does not have the cash on hand to make the investment?
  - Could Baxter raise $100,000 in new equity to make the investment?

Agency costs of borrowing: Debt Overhang and Under-Investment

	Without New Project	With New Project
Existing assets	900	900
New project		150
Total firm value	900	1050
Debt	900	1000
Equity	0	50

Cashing out

When a firm faces financial distress:

- Pay an immediate cash dividend to the shareholders
  - Sell assets below market value
  - Borrow more (leverage ratchet effect)

Agency costs of borrowing

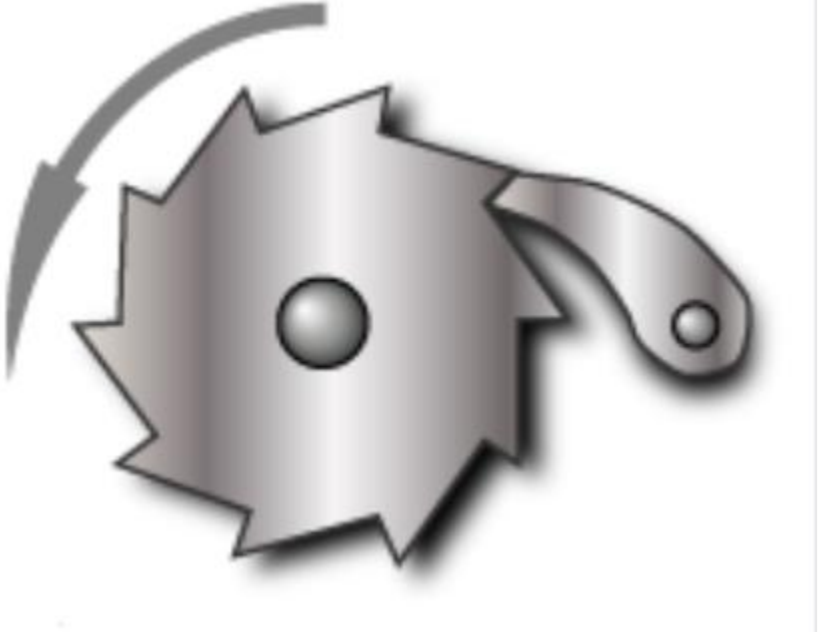
- Agency costs of borrowing include:

- Higher rates on debt due to bondholder’s expectation that equity investor’s interest will be prioritised
- Restrictive covenants in debt contracts give rise to:
  - Direct cost of monitoring them,
    and
  - Indirect cost of lost investment opportunities

Agency costs of borrowing: The Leverage Ratchet Effect

Once existing debt is in place:

1. Shareholders may have an incentive to increase leverage even if it decreases the value of the firm, and
2. Shareholders will not have an incentive to decrease leverage by buying back debt, even if it will increase the value of the firm



Agency costs of borrowing: The Leverage Ratchet Effect

- Baxter's shareholders would not gain by reducing leverage from $1 million to $400,000, even though firm value would increase by eliminating the cost of underinvestment.
- the debt will be default-free after the buyback,
- New equity: $\$600{,}000/1.05 = \$571{,}429$
- raise another $100,000 for the new project, 50% return rate
- 900+150=1050
- 1050-400=650<571

Improve conflicts of interest among interest groups (agency)

1. Disciplining by debt
Free Cash Flows Hypothesis (Michael Jensen, AER 1976)

Managers may have incentives to make 'wasteful' use of a firm's free cash flows (i.e., cash flows from operations over which they have discretionary spending power)

Borrowing introduces discipline since it creates the commitment to make future payments increases the risk of default on projects with sub-standard returns

Improve conflicts of interest among interest groups (agency)

2. Monitoring Nature of Debt → Signalling

Forgiving nature of equity commitment

VERSUS

Inflexibility of debt commitment

Other costs of debt

Loss of Flexibility

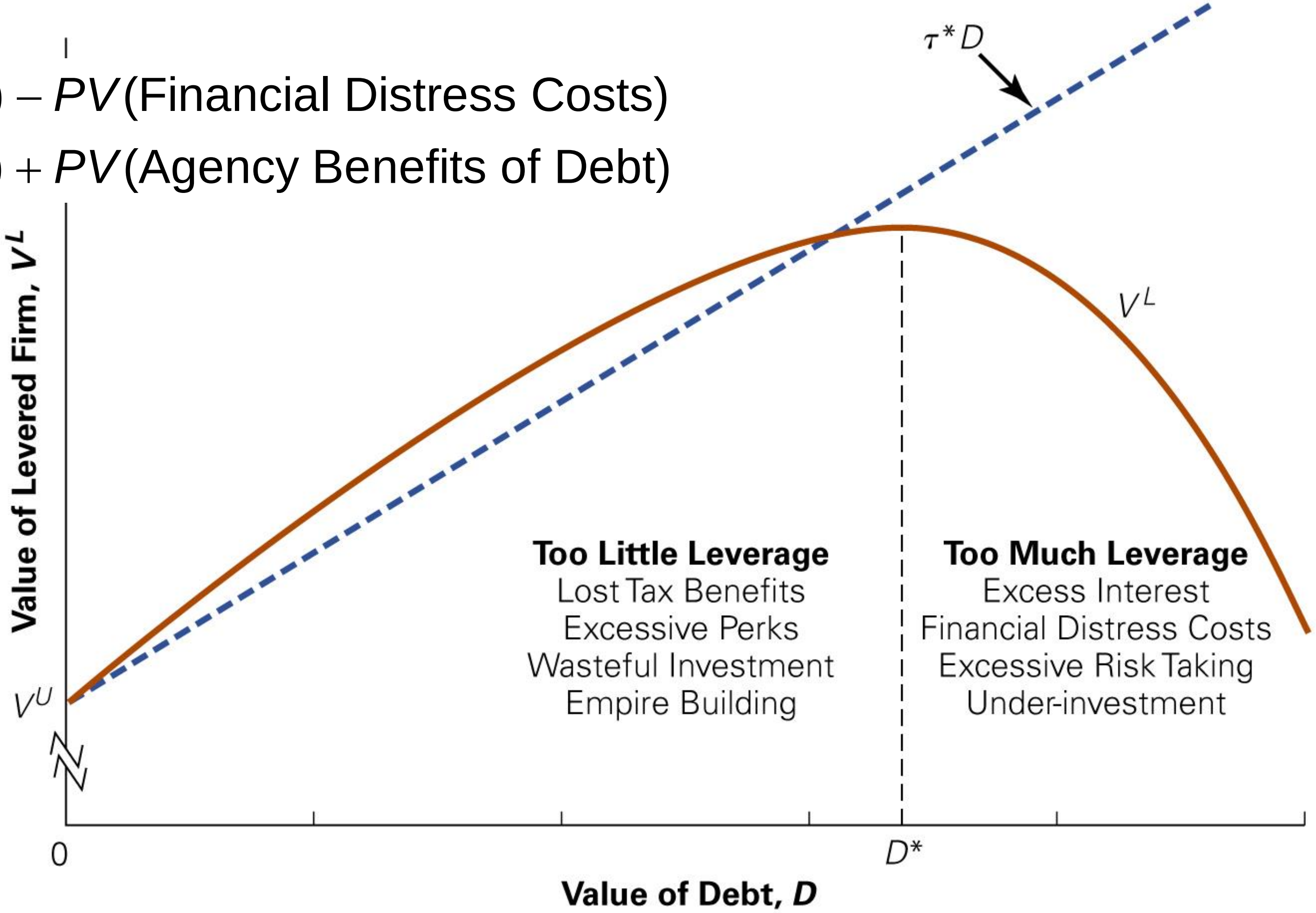
Due to:

(i) commitment to future payments
(ii) lenders protecting their interests against equity investors' actions

Value of financial flexibility
=
Cost of maintaining large cash balances and excess debt capacity

Trade-off theory

$$V^L = V^U + PV(\text{Interest Tax Shield}) - PV(\text{Financial Distress Costs})$$
$$- PV(\text{Agency Costs of Debt}) + PV(\text{Agency Benefits of Debt})$$



Dynamic capital structure

- Managers do not optimise capital structure period by period, but as the result of a dynamic process that accounts for the costs associated with capital structure adjustments

The Pecking Order of financing opportunities:

1. Retained earnings are preferred to external sources of funds ⇒ dividend policy reflects the difference between earnings and investment needs; i.e., dividends are a residual

2. If there is excess cash, debt is paid off before shares are repurchased.

3. If external financing is needed, firms issue the safest security first; i.e., straight debt first, then convertible bonds, then equity

Pecking order:

A hierarchy of status seen among members of a group of people or animals, originally as observed among hens

Pecking order theory: Issuing Equity and Adverse Selection

- Adverse Selection

  The idea that when the buyers and sellers have different information, the average quality of assets in the market will differ from the average quality overall

- Lemons Principle

  When a seller has private information about the value of a good, buyers will discount the price they are willing to pay due to adverse selection

Pecking order theory

Pecking order theory: capital structure is determined by the drive to find the less expensive way to finance new investment

→ retained earnings first, then lower-risk debt, all the way to equity

→ no well-defined target debt ratio

Empirical implications of pecking order

- **Negative relationship between profitability and leverage**
- **Managers more likely to issue shares when fairly or over-priced → share value falls on the announcement of new equity issue**
- **Financing by retained earnings or debt does not convey information → no effect on share price**
- **Since issuing shares is 'bad news', firms may under-invest rather than raise equity, or accept 'excessive' leverage**
- **Under-investment less severe after disclosure of information → share issue cluster around such disclosures**
- **Firms with less tangible assets are more subject to informational asymmetries**
  →under-investment is more severe
  →more debt
- **Financial slack is valuable**

Thank you
for your attention!

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